

NAME _____



GOSFORD HIGH SCHOOL

2016

Higher School Certificate

MATHEMATICS

Assessment Task 2

Time Allowed – 90 minutes

General Instructions

- Reading time 5 minutes
- Working time 90 minutes
- Write using black or blue pen
- Board approved calculators may be used
- For all questions show relevant mathematical reasoning and/or calculations
- Use the attached multiple choice answer sheet to answer questions 1 to 5
- Start each question from 6 to 9 on a new page
- Total marks 65

MULTIPLE CHOICE

5 marks

1 mark each

Question 1

The equation $(\cos\theta - 1)(2\sin\theta - 1) = 0$ for $0 \leq \theta \leq 2\pi$ has

- A) 2 solutions
- B) 3 solutions
- C) 4 solutions
- D) 5 solutions

Question 2

$$\int \frac{1}{\sqrt{1-4x}} dx =$$

- A) $-\frac{\sqrt{1-4x}}{2} + c$
- B) $-\frac{\sqrt{(1-4x)^3}}{2} + c$
- C) $2\sqrt{1-4x} + c$
- D) $\frac{2\sqrt{(1-4x)^3}}{3} + c$

Question 3

Find the range of values of the function $y = f(x)$,

given that $f(x) = 4 - x^2$ in the domain $-2 \leq x \leq 4$

- A) $y \leq 4$
- B) $0 \leq y \leq 4$
- C) $-12 \leq y \leq 0$
- D) $-12 \leq y \leq 4$

Question 4

Given that $\frac{dy}{dx} = \cos x - \sin 2x$ and that when $x = 0, y = 1.5$, then

- A) $y = -\sin x - \frac{1}{2}\cos 2x + 2$
- B) $y = \sin x + \frac{1}{2}\cos 2x + 1$
- C) $y = \sin x - 2\cos 2x + 3.5$
- D) $y = -\sin x + 2\cos 2x - 0.5$

Question 5

If $\frac{d^2y}{dx^2} = (x + 2)^2(x - 1)$, then point(s) of inflexion occur at

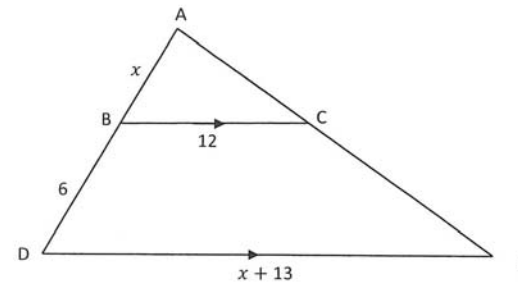
- A) $x = -2$ only
- B) $x = -2$ and $x = 1$
- C) $x = 1$ only
- D) neither $x = -2$ or $x = 1$

Question 6

15 marks

- (a) Factorise $8 + m^3$ (1)
- (b) Solve $|x - 3| = 2x$ (2)
- (c) Find the vertex and the focus of the parabola $x^2 = 8y - 16$ (2)
- (d) Find the perpendicular distance from the point $(1, -2)$ to the line with equation $3x + 4y - 6 = 0$ (2)
- (e) A triangle has sides of $AC = 4\text{cm}$, $AB = 5\text{cm}$ and $BC = 6\text{cm}$. Find angle BAC, correct to the nearest degree. (2)

(f)



- (i) Prove $\triangle ABC$ is similar to $\triangle ADE$ (2)
- (ii) Using the dimensions as shown on the diagram, find x (2)
- (g) For what values of k is the quadratic expression $x^2 - 3kx + 9k$ positive definite. (2)

Question 7

15 marks

- (a) Copy and complete the following function table for $f(x) = \frac{4}{x+1}$.

X	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
$f(x)$	4		$\frac{8}{3}$	$\frac{16}{7}$	2

(1)

If $J = \int_0^1 \frac{4}{x+1} dx$,

find the value of J using Simpson's rule and 5 function values

(3)

- (b) Find $\int x\sqrt{x} dx$ (1)

- (c) Evaluate $\int_{-1}^1 (x^2 + 1)^2 dx$ (2)

- (d) The two curves $y = x^2 + 3$ and $y = x^3 - 1$ intersect at the point $(2, 7)$.

- (i) Draw the two curves on the same number plane and label the given point of intersection. (2)

- (ii) Find the area of the region enclosed by the two curves and the y axis. (3)

- (e) The arc of the curve $y = \frac{4}{x}$ between the points $(1, 4)$ and $(4, 1)$ is rotated about the y axis. Find the volume of the solid generated. (3)

Question 8

15 marks

- (a) State the period and amplitude of the curve $y = 3 + \cos\left(\frac{x}{2}\right)$ (2)

- (b) What is the exact value of $\sin\left(\frac{4\pi}{3}\right)$ (1)

- (c) The area of a sector AOB of a circle, with centre O and radius 6cm, is 27cm^2 .

- i) Calculate the angle AOB in radians (1)

- ii) Find the length of the minor arc AB (1)

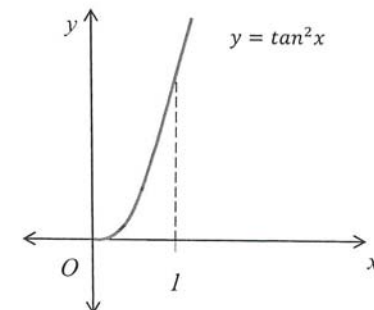
- (d) Find $\frac{d}{dx}[\tan^2 2x]$ (1)

- (e) The curve $y = \sqrt{\cos \pi x}$, where $0 \leq x \leq \frac{1}{2}$ is rotated about the x axis.

What is the volume of the solid of revolution thus generated? (3)

- (f) What is the gradient of the tangent to the curve $y = \frac{x}{\sin 2x}$ at the point on the curve where $x = \frac{\pi}{4}$ (2)

- (g) A portion of the graph of $y = \tan^2 x$ is shown below. Find the area bounded by the curve $y = \tan^2 x$, the x axis and the ordinate at $x = 1$. (Write your answer correct to 3 significant figures) (4)



Question 9

15 marks

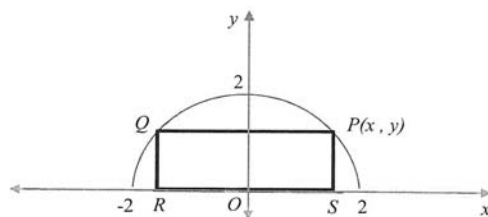
MULTIPLE CHOICE ANSWER SHEET

STUDENT NAME/NUMBER _____

(a) Find the second derivative of $(x^2 + 1)^5$ (2)(b) (i) Find the co-ordinates of the stationary points on the curve
 $y = x^3 - 6x^2 + 9x - 4$ (3)

(ii) Determine the nature of these stationary points (2)

(iii) Sketch the curve (2)

(c) A rectangle $PQRS$ is inscribed inside the semi-circle $y = \sqrt{4 - x^2}$ as shown in the diagram below. P and Q lie on the semi-circle.(i) If P has co-ordinates (x, y) , show that the area of the rectangle (A) is given by $A = 2x(4 - x^2)^{\frac{1}{2}}$. (1)(ii) Show that $\frac{dA}{dx} = \frac{4(2 - x^2)}{\sqrt{4 - x^2}}$. (2)(iii) Hence prove that the area of the rectangle is a maximum when $x = \sqrt{2}$. (2)

(iv) Find this maximum area (1)

1. A B C D

2. A B C D

3. A B C D

4. A B C D

5. A B C D

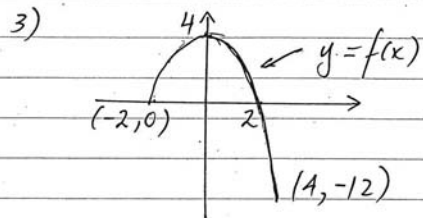
Multiple Choice

1) $\cos \theta = 1$ or $\sin \theta = \frac{1}{2}$
 $\therefore \theta = 0, 2\pi$ $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

$\therefore$ 4 solutions (C)

2) $\int (1-4x)^{-\frac{1}{2}} dx = \frac{(1-4x)^{\frac{1}{2}}}{\frac{1}{2} \times (-4)} + C$
 $= \frac{\sqrt{1-4x} + C}{-2}$

$\therefore$ (A)



Range is $-12 \leq y \leq 4$ (D)

4) $y = \sin x + \frac{1}{2} \cos 2x + C$
 when $x = 0, y = 1.5$
 $1.5 = \sin 0 + \frac{1}{2} \cos 0 + C$
 $1.5 = 0.5 + C$
 $1 = C$

$\therefore y = \sin x + \frac{1}{2} \cos 2x + 1$ (B)

5) $\frac{d^2y}{dx^2} = 0$ at $x = -2, 1$

when $x < -2$ $\frac{d^2y}{dx^2} < 0$
 (say $x = -3$) $\frac{d^2y}{dx^2} < 0$

when $-2 < x < 1$ $\frac{d^2y}{dx^2} < 0$
 (say $x = 0$) $\frac{d^2y}{dx^2} < 0$

when $x > 1$ $\frac{d^2y}{dx^2} > 0$
 (say $x = 2$) $\frac{d^2y}{dx^2} > 0$

$\therefore$ Pt. of inflexion at (C)
 $x = 1$ only

Question 6 (1 mark)

a) $8 + m^3 = (2+m)(4-2m+m^2)$

b) $x - 3 = 2x$
 $-3 = x$

But this value does not satisfy since $6 \neq -6$.

OR $-(x-3) = 2x$
 $-x + 3 = 2x$

$3 = 3x$ 1 mark
 $1 = x$ for 2 answers

Which satisfies
 since $2 = 2$

$\therefore x = 1$ only 1 mark

Question 6 (continued)

c) $x^2 = 8(y-2)$

$\therefore 4a = 8 \rightarrow a = 2$

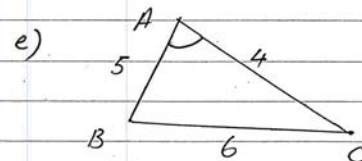
Vertex is $(0, 2)$ 1 mark

$\therefore$ Focus is $(0, 4)$ 1 mark

d) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$= \frac{|1(3) + 4(-2) - 6|}{\sqrt{3^2 + 4^2}}$ 1 mark

$d = \frac{11}{5}$ 1 mark for answer



$\cos A = \frac{5^2 + 4^2 - 6^2}{(2 \times 5 \times 4)}$

$= \frac{5}{40}$
 $= \frac{1}{8}$ 1 mark

$A = \cos^{-1}\left(\frac{1}{8}\right)$

$A = 83^\circ$ 1 mark

f) i) In Δ 's ABC & ADC

$\hat{A}$ is common

$\hat{ABC} = \hat{ADC}$ (corresponding

angles, $BC \parallel DE$) 1 mark.
 $\therefore \Delta ABC \parallel \Delta ADE$ (triangles are equiangular) 1 mark.

ii) $\frac{x}{x+6} = \frac{12}{x+13}$ 1 mark

(corresponding sides of similar triangles are in proportion)

$x(x+13) = 12(x+6)$

$x^2 + 13x = 12x + 72$

$x^2 + x - 72 = 0$

$(x+9)(x-8) = 0$ 1 mark

$\therefore x = 8$ only

since $x > 0$

g) Positive Definite

if $\Delta < 0$ noting that coefficient of $x^2 > 0$

$\therefore b^2 - 4ac < 0$

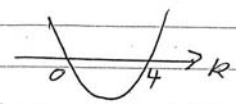
$(-3k)^2 - 4(1)(9k) < 0$

$9k^2 - 36k < 0$ 1 mark

$9k(k-4) < 0$

$\therefore 0 < k < 4$

1 mark



Question 7

a)	x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
	$f(x)$	4	3.2	$\frac{8}{3}$	$\frac{16}{7}$	2

1 mark

$$J \doteq \frac{h}{3} \left[f(0) + f(1) + 2f\left(\frac{1}{2}\right) + 4 \left[f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) \right] \right]$$

1 mark for ^{correct} knowledge of Simpson's Rule.

$$\doteq \frac{0.25}{3} \left[4 + 2 + 2 \times \frac{8}{3} + 4 \left(\frac{16}{5} + \frac{16}{7} \right) \right]$$

1 mark.

$$\doteq \frac{17.47}{630} \approx 2.8 \text{ (to 1 d.p.)}$$

1 mark (rounding permitted)

$$b) \int x\sqrt{x} \, dx = \int x^{\frac{3}{2}} \, dx.$$

1 mark

$$= \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + c.$$

$$= \frac{2x^{\frac{5}{2}}}{5} + c$$

1 mark.

$$= \frac{2x^2\sqrt{x}}{5} + c$$

$$c) \int_{-1}^1 (x^2+1)^2 \, dx = 2 \int_0^1 (x^4 + 2x^2 + 1) \, dx.$$

$$= 2 \left[\frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^1$$

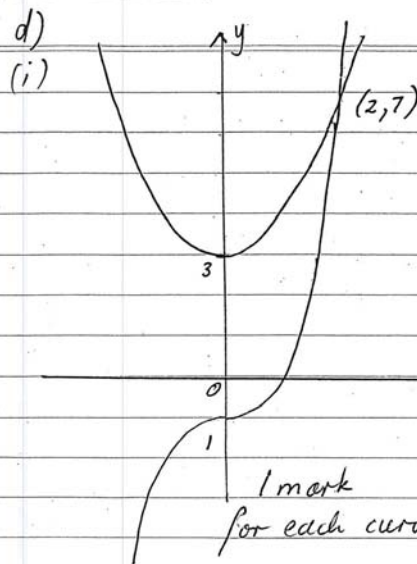
1 mark.

$$= 2 \left[\frac{1}{5} + \frac{2}{3} + 1 \right] = \frac{56}{15}$$

$$= 3.73$$

1 mark.

Question 7 (continued)



$$(ii) \text{ Area} = \int_0^2 (x^2+3) - (x^3-1) \, dx.$$

1 mark.

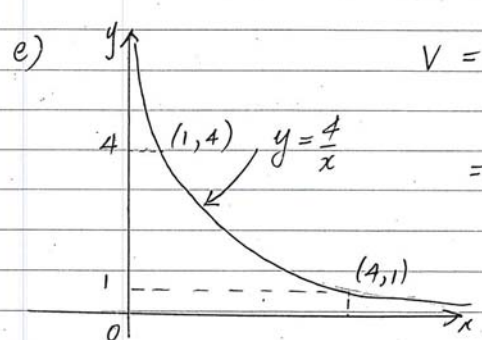
$$= \int_0^2 (x^2 - x^3 + 4) \, dx.$$

$$= \left[\frac{x^3}{3} - \frac{x^4}{4} + 4x \right]_0^2$$

1 mark.

$$= \left[\frac{8}{3} - 4 + 8 - 0 \right]$$

1 mark for each curve. = $6\frac{2}{3}$ sq. units. 1 mark.



$$V = \pi \int_1^4 \left(\frac{4}{y} \right)^2 \, dy. \text{ since } x = \frac{4}{y}.$$

1 mark.

$$= \pi \int_1^4 16y^{-2} \, dy.$$

$$= 16\pi \left[\frac{y^{-1}}{-1} \right]_1^4$$

1 mark.

$$= -16\pi \left[\frac{1}{4} - 1 \right]$$

$$= -16\pi \times \left(-\frac{3}{4} \right)$$

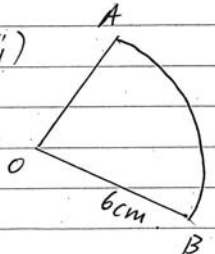
1 mark.

$$= 12\pi \text{ cubic units}$$

Question 8

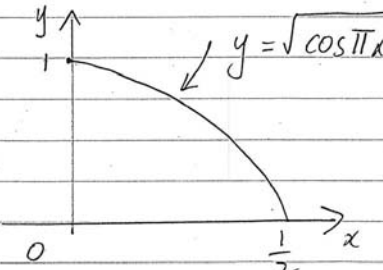
a) Period = $\frac{2\pi}{1/2}$ Amplitude = 1
 $= 4\pi$ (1 mark) (1 mark)

b) $\sin\left(\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}$ (1 mark)

c) (i)  $A = 27 \text{ sq cm.}$
 $\therefore \frac{1}{2} r^2 \theta = 27.$
 $\frac{1}{2} (6)^2 \theta = 27.$
 $\theta = \frac{27}{18}.$
 $\theta = \frac{3}{2} \text{ radians. (1 mark)}$

(ii) $L_{AB} = r\theta.$
 $= 6 \times \frac{3}{2}.$
 $= 9 \text{ cm. (1 mark)}$

d) $\frac{d}{dx} \left[(\tan 2x)^2 \right] = 2(\tan 2x) \times 2 \sec^2 2x. \quad (1 \text{ mark})$
 $= 4 \tan 2x \sec^2 2x.$

e)  $y = \sqrt{\cos \pi x}.$
 $V = \pi \int_0^{1/2} \cos \pi x \, dx. \quad (1 \text{ mark})$
 $= \pi \left[\frac{1}{\pi} \sin \pi x \right]_0^{1/2} \quad (1 \text{ mark})$
 $= \sin \frac{\pi}{2} - \sin 0$
 $= 1 \text{ sq unit. (1 mark)}$

Question 8 (continued)

f) $y = \frac{x}{\sin 2x}$

$\frac{dy}{dx} = \frac{\sin 2x \cdot 1 + x \cdot 2 \cos 2x}{\sin^2 2x} \quad (1 \text{ mark})$
 $= \frac{\sin 2x + 2x \cos 2x}{\sin^2 2x}.$

$= \frac{1 + \frac{\pi}{2} \times 0}{1} \quad \text{at } x = \frac{\pi}{4}$

$= 1 \quad (1 \text{ mark})$

$\therefore$ Gradient of required tangent is 1

g) Area = $\int_0^1 \tan^2 x \, dx. \quad (1 \text{ mark})$

$= \int_0^1 (\sec^2 x - 1) \, dx. \quad (1 \text{ mark})$

$= [\tan x - x]_0^1 \quad (1 \text{ mark})$

$= \tan 1 - 1$

$= 0.557 \quad (1 \text{ mark})$

Question 9

a) Let $y = (x^2 + 1)^5$

$$\frac{dy}{dx} = 5(x^2 + 1)^4 \times 2x = 10x(x^2 + 1)^4 \quad (1 \text{ mark})$$

$$\frac{d^2y}{dx^2} = (x^2 + 1)^4 \times 10 + 10x \times 4(x^2 + 1)^3 \times 2x$$

$$= 10(x^2 + 1)^4 + 80x^2(x^2 + 1)^3$$

$$= 10(x^2 + 1)^3 [(x^2 + 1) + 8x^2] \quad (1 \text{ mark})$$

$$= 10(x^2 + 1)^3 (9x^2 + 1)$$

b) (i) $y = x^3 - 6x^2 + 9x - 4$

$$\frac{dy}{dx} = 3x^2 - 12x + 9$$

$$\frac{d^2y}{dx^2} = 6x - 12$$

For stationary points $\frac{dy}{dx} = 0$

$$\therefore 3x^2 - 12x + 9 = 0 \quad (1 \text{ mark})$$

$$x^2 - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0 \quad (1 \text{ mark})$$

$$x = 1, 3$$

$\therefore$ Stationary points at $(1, 0) \neq (3, -4)$ (1 mark)

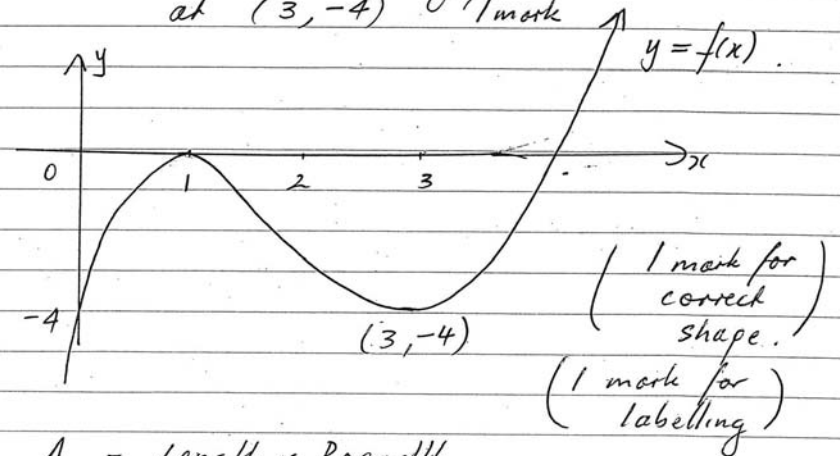
(ii) $\frac{d^2y}{dx^2} = -6 < 0$ when $x = 1$

$\therefore$ A maximum turning point exists at $(1, 0)$ (1 mark)

Question 9 (b) (continued)

$$\frac{d^2y}{dx^2} = 6 > 0 \quad \text{when } x = 3$$

$\therefore$ A minimum turning point exists at $(3, -4)$ 1 mark



(c) (i) $A = \text{Length} \times \text{Breadth}$

$$= 2x \times y$$

$$A = 2x \sqrt{4 - x^2}$$

$$A = 2x(4 - x^2)^{\frac{1}{2}} \quad (1 \text{ mark})$$

(ii) $\frac{dA}{dx} = (4 - x^2)^{\frac{1}{2}} \times 2 + 2x \times \frac{1}{2} (4 - x^2)^{-\frac{1}{2}} \times (-2x)$

$$= 2(4 - x^2)^{\frac{1}{2}} - 2x^2(4 - x^2)^{-\frac{1}{2}} \quad (1 \text{ mark})$$

$$= 2(4 - x^2)^{-\frac{1}{2}} [4 - x^2 - x^2]$$

$$= 2(4 - x^2)^{-\frac{1}{2}} (4 - 2x^2) \quad (1 \text{ mark for show})$$

$$= \frac{4(2 - x^2)}{\sqrt{4 - x^2}} \quad \text{as required.}$$

Question 7(c) (continued)

(iii) For stationary points $\frac{dA}{dx} = 0$.

$$\therefore 2 - x^2 = 0$$
$$x = \sqrt{2} \text{ since } x > 0 \quad (1 \text{ mark})$$

when $x < \sqrt{2}$ $\frac{dA}{dx} > 0 \therefore$ increasing.
(say $x = 1$)

when $x > \sqrt{2}$ $\frac{dA}{dx} < 0 \therefore$ decreasing.
(say $x = 1.5$)

$\therefore$ A maximum value of A occurs
when $x = \sqrt{2}$ (1 mark)

$$(iv) \text{ Max. Area} = 2 \times \sqrt{2} (4 - (\sqrt{2})^2)^{\frac{1}{2}}$$

$$= 2\sqrt{2} \times \sqrt{2}$$

$$= 4 \text{ square units} \quad (1 \text{ mark})$$